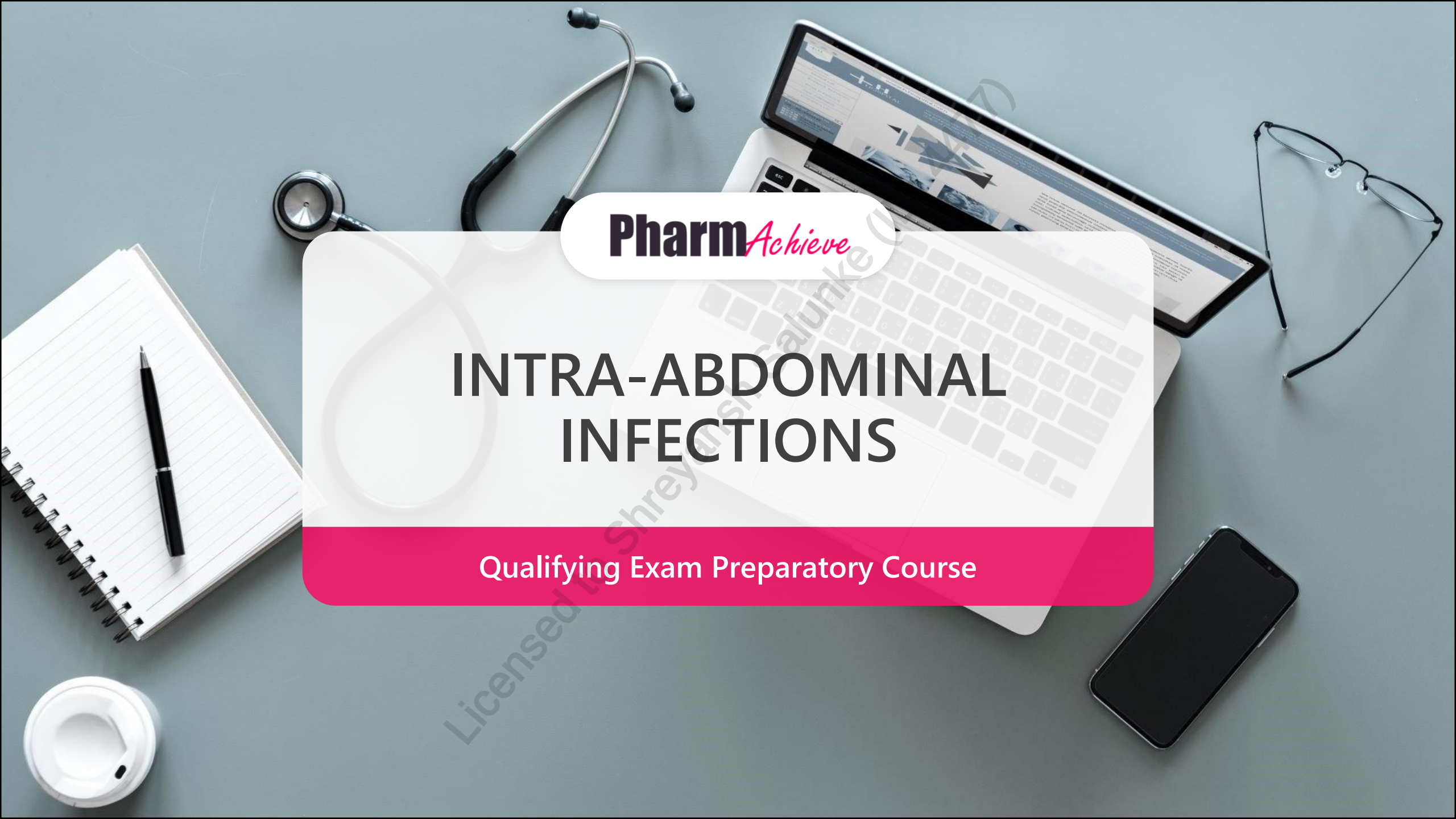


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

INTRA-ABDOMINAL INFECTIONS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

Abx Antibiotic

ALP Alkaline Phosphatase

AST/ALT Aspartate Transaminase/Alanine Transaminase

BUN Blood Urea Nitrogen

CO Carbon Monoxide

CT Computer Tomography

ESBL Extended-Spectrum Beta-lactam

FQ Fluoroquinolone

GI(T) Gastrointestinal (Tract)

Gen Generation

GERD Gastroesophageal Reflux Disease

HCT Hematocrit

IM Intramuscular

IV Intravenous

MDRO Multi Drug-Resistant Organism

MRSA Methicillin-Resistant Staphylococcus Aureus

N/V Nausea/Vomiting

Pip-Taz Piperacillin-Tazobactam

SBP Spontaneous Bacterial Peritonitis

WBC White Blood Cells

DEFINITIONS AND COMMON PATHOGENS

Intra-abdominal abscess

- Pocket of infected fluid and pus located inside the abdominal cavity and surrounded by inflamed tissue

Peritonitis

- Inflammation of the peritoneum, the thin tissue that covers the inner wall of the abdomen and abdominal organs
- Usually *E.coli* (40-60%) or *K. pneumoniae* (primary) or mixed (if secondary)

Cholangitis

- Inflammation or infection of the bile duct/biliary tree Most commonly gram-negative: *E. coli*, *Klebsiella pneumonia*, *Enterobacter spp.*, *Proteus spp.*

Cholecystitis

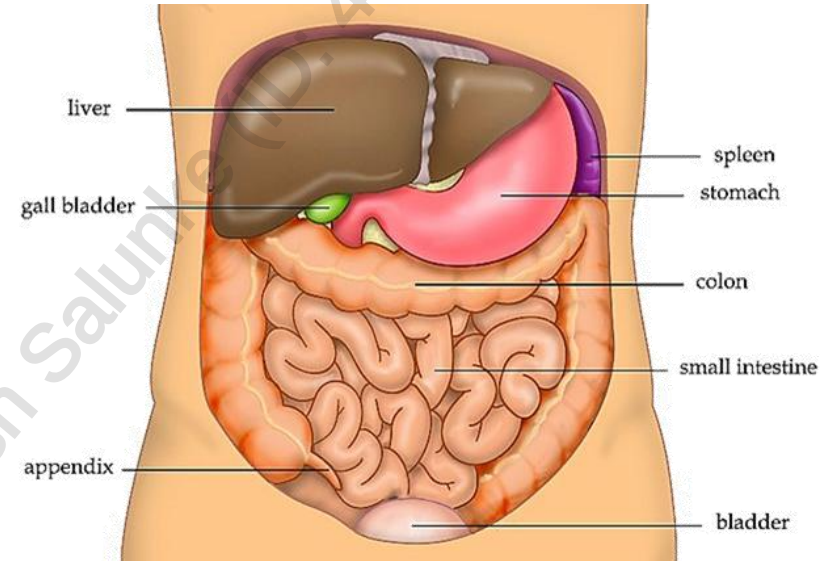
- Inflammation of the gallbladder usually due to obstruction
- Most commonly gram-negative: *E. coli*, *Klebsiella pneumonia*, *Enterobacter spp.*, *Proteus spp.*

Acute appendicitis

- Inflammation of the vermiform appendix due to obstruction
- *E. coli*, *Enterococcus*, *Viridans streptococci*, *Bacteroides*, *Lactobacillus*, *Peptostreptococcus*

PATHOPHYSIOLOGY AND RISK FACTORS

Risk factors: GI malignancy, trauma, biliary stone, surgical biliary intervention, geriatrics, cirrhosis, bleeding/obstructing/perforating GI ulcers (serious condition requiring surgery where untreated ulcer can burn through GIT wall, allowing digestive contents to enter the abdominal cavity, causing infection and inflammation)



Primary peritonitis

- The cause can be exogenous (e.g., catheter) or endogenous (e.g., cirrhotic ascites)
- Often occurs in patients with ascites due to cirrhosis

Secondary peritonitis

- Occurs because of a primary infectious process or macroscopic spillage of GI contents into the peritoneal cavity

CLINICAL PRESENTATION

Primary peritonitis (largely monomicrobial)	Secondary peritonitis (polymicrobial)
• Fever (most common sign) • Abdominal pain • Abdominal tenderness • Hypotension • 10% of cases may show no symptoms (silent)	• Abdominal pain • Abdominal distension • Fever • Hypotension

CLINICAL PRESENTATION

Cholangitis/Cholecystitis	Acute appendicitis
• N/V • Right upper quadrant pain • Fever (cholangitis>cholecystitis) • Hypotension (cholangitis>cholecystitis)	• Early symptoms: indigestion, flatulence, diarrhea or constipation, nonspecific abdominal pain • Late symptoms: right lower quadrant pain, nausea/vomiting, fever (high if ruptured, increased WBC)

COMPLICATIONS AND DIAGNOSIS

If peritonitis is left untreated, the following complications may occur:

- Bacterial proliferation, which stimulates immunity and inflammation
- Fluid shift to peritoneum: hypotension, renal failure, electrolyte imbalances, acidosis, sepsis, shock and death, 3rd spacing, ↓ blood volume and cardiac output

Condition	Diagnosis
Cholecystitis	• Ultrasonography reveal distention of gallbladder, edema, stones
Cholangitis	• CT/MRI imaging reveal extent of biliary obstruction
Peritonitis	• Clinical symptoms (ex. rebound tenderness) • Paracentesis tests ascitic fluid for elevated polymorphonuclear count (if > 250 cells/mm³ with (+) culture → diagnosis of primary peritonitis)

GOALS OF THERAPY

- ✓ Eradicate infective pathogens and cure the infection
- ✓ Achieve source control, if applicable
- ✓ Relieve the signs and symptoms and normalize lab values
- ✓ Prevent complications: secondary infections, intra-abdominal abscess formation, sepsis, multiple organ failure, death
- ✓ Prevent recurrence of infection (i.e., SBP)

NON-PHARMACOLOGICAL MEASURES

1. Source control including physical measures and surgery (urgent if vitals are unstable)
 - Abscess drainage, appendix removal, debridement of necrotic tissue, washout of peritoneum
2. Supportive measures
 - Fluid, electrolyte supplementation and analgesia for pain management post surgery

PERITONITIS

Intra-abdominal Infections	Common Pathogens	Empirical Pharmacological Therapy
Primary Peritonitis Community-acquired (without risk of resistance or failure)	Monomicrobial • <i>E. coli</i> • <i>Klebsiella pneumonia</i> • <i>S. pneumoniae</i> or other streptococci	• 3rd gen cephalosporin (e.g., cefotaxime, ceftriaxone) • FQ • Carbapenem (e.g. Ertapenem)
Secondary Peritonitis Community-Associated Health Care-Associated High Risk*	• Enterobacteriales • Anaerobes • Enterococcus • Yeast • MDROs	• 1st/2nd/3rd gen ceph + metronidazole • Amoxicillin-clavulanate • Piperacillin-tazobactam • Carbapenem • Aminoglycosides + metronidazole • FQ +/- metronidazole

*High Risk describes patients with ↑risk of treatment failure and/or resistance

- Delay in initial intervention (>24 hours)
- High severity of illness (APACHE II score ≥ 15)
- Diffuse peritonitis or degree of peritoneal involvement
- Advanced age (>70 years)
- Low albumin or poor nutrition
- Comorbidity or organ dysfunction
- Inability to achieve adequate debridement
- Presence of malignancy

Duration of therapy varies from 4-∞ days depending on patient's clinical response and adequacy of source control

OTHER THERAPIES

Intra-abdominal Infections	Common Pathogens	Empirical Pharmacological Therapy
Cholangitis Hydration and biliary decompression required	• <i>E. coli</i>, <i>Klebsiella</i>, <i>Enterococcus</i> species - less commonly anaerobes	• 2nd or 3rd generation cephalosporin +/- Metronidazole • Amoxicillin-clavulanate • Piperacillin-Tazobactam • Meropenem, Imipenem or Ertapenem
Cholecystitis Surgery required but not generally emergent	• <i>E. coli</i>, <i>Klebsiella</i>, <i>Enterococcus</i> species and anaerobes	• Same as cholangitis • Surgery done as elective procedure after improvement in infection
Acute appendicitis Surgery OR antibiotics	• <i>E. coli</i>, <i>Enterococcus</i>, anaerobes	• 1st/2nd/3rd gen cephalosporin + metronidazole • Do not need more than 1-3 doses unless gangrenous or perforated appendicitis

OTHER THERAPIES

Intra-abdominal Infections	Common Pathogens	Empirical Pharmacological Therapy
Abscess Surgical debridement and drainage of abscess with IV abx	• Gut anaerobes (B. fragilis) and gram negative bacteria • Candida (less likely)	• Ceftriaxone + Metronidazole • Ciprofloxacin + Metronidazole • Carbapenem • Piperacillin-tazobactam • Antifungals if needed
Acute pancreatitis	• Non-infectious	• Supportive care with fluid resuscitation, pain control and nutritional support • Prophylactic abx are NOT recommended regardless of type and severity

ANTIBIOTIC THERAPY: APPENDICITIS

- Antibiotics +/- surgery for uncomplicated infection
- Surgery for gangrenous or perforated appendicitis
- Should be directed at both G⁻ aerobes and anaerobes
- Duration of therapy:
 - If uncomplicated (non-perforated): Single dose pre-operatively
 - If gangrenous or perforated appendicitis: 4- ∞ days (dependent on source control)
- Combo therapy of antimicrobials (metronidazole + aminoglycoside) was shown to be as efficacious as monotherapy (beta-lactam/beta-lactam inhibitor, carbapenems)
- Preferred choice is 1st/2nd/3rd generation cephalosporin: (ex. cefazolin, cefotetan, cefoxitin or ceftriaxone) + metronidazole pre-op
- If undergoing surgery, do not need post operative antimicrobial unless gangrenous or perforated appendicitis
- If using antimicrobials alone for uncomplicated appendicitis, 10 days of therapy (FQ + metronidazole)

FACTORS TO CONSIDER

1. Start IV antibiotic empirically based on suspected pathogen
2. IV administration – *ensures adequate systemic and tissue concentrations, especially in patients in shock*
 - Due to poor perfusion of the muscle or GIT, oral or IM routes are not recommended
3. Enterococcus suspicions (rarely requires directed therapy unless grown in abscess/blood)

Enterococcus faecalis:

- Ampicillin
- *Alternative treatment:* Vancomycin, Daptomycin, Linezolid

Enterococcus faecium:

- Vancomycin
- *Alternative treatment:* Daptomycin, Linezolid

4. Antifungal coverage:

- Consider if patient is immunocompromised, or if hospital acquired infection
- Positive blood culture for Candida

MONITORING PARAMETERS

Efficacy endpoints		
Parameter	Degree of Change	Timeframe* *Varies based on condition, severity and treatment (including source control)
Nonspecific symptoms (fever, chills, nausea/vomiting)	Significant improvement or resolution	Within 2-4 days
Abdominal pain/tenderness/distension, WBC normalization		Follow up on symptoms in 24 hours. If no clinical response in a few days, re-assess for possible causes of treatment failure
Secondary infections, abscess formation, sepsis, multiple organ failure, death	Prevent	Duration of therapy

Safety endpoints		
Parameter	Degree of Change	Timeframe
Adverse effects (nausea/vomiting, diarrhea)	Prevent or minimize	Duration of therapy

CASE 1

JC, a 52-year-old male, was brought in by his son to the ER for abdominal pain. JC's son reports JC has been experiencing severe abdominal pain that impairs his ability to complete his daily activities. In addition to his abdominal pain, JC reports nausea and vomiting and has had minimal oral intake over the past several days. JC's medical history is significant for cirrhosis with a history of ascites, GERD, seasonal allergies, and a chronic hepatitis C infection. He's currently taking spironolactone 100 mg PO once daily and furosemide 40 mg PO daily. He has a history of alcohol abuse and has been drinking 10 beers a night since he was 20. Today, he is febrile (temperature 38.5°C, he has no hematemesis or melena, his BP is 145/80 mmHg, his pulse is 100 bpm, and his RR is 30 bpm). He is also jaundiced and has spider angiomas on his abdomen.

He has no allergies and no recent antimicrobial use.

1. What pathogen is unlikely to cause primary peritonitis in this case?
 - a) *Streptococcus agalactiae*
 - b) *E. coli*
 - c) *Pseudomonas aeruginosa*
 - d) *Klebsiella pneumoniae*

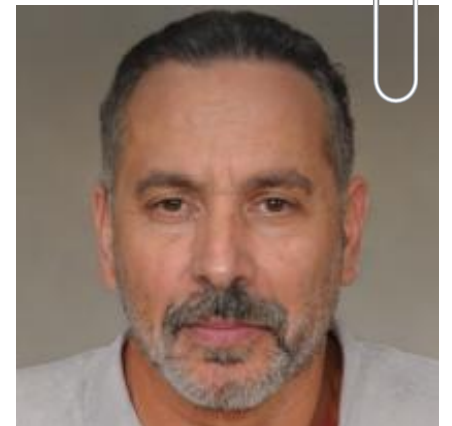


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2. What regimen is preferred empirically for JC's treatment?
- a) Norfloxacin
 - b) Ceftriaxone
 - c) Vancomycin
 - d) Meropenem



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3. JC undergoes paracentesis and grows *Klebsiella pneumoniae* susceptible to all tested antimicrobials. What should be JC's therapy?
- a) Amoxicillin
 - b) Cefazolin
 - c) Moxifloxacin
 - d) Piperacillin/tazobactam

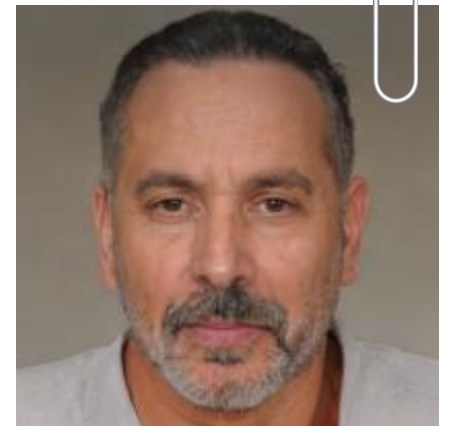


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4. How many days of total antimicrobial therapy should JC receive antimicrobial therapy for?
- a) 3 days
 - b) 5 days
 - c) 10 days
 - d) 14 days



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1. Chow AW et al. *Canadian practice guidelines for surgical intra-abdominal infections*. Can J Infect Dis Med Microbiol. 2010;21(1):11-37.
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CHANGE LOG

March 2025

- 2024 IDSA guidelines added
- Editorial changes throughout

July 2025

- Slides 5, 6, 7: Updated and simplified information
- Case updated

September 2025

- Slide 16: Elaborated on monitoring parameters

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 3: Updated legend
- Slide 16: Updated monitoring parameters